

Build Instructions — Windows Standalone (.exe)

Step-by-step guide to produce a runnable `Space Shooter.exe`. Assumes you have already opened the project and run **Space Shooter ▶ Setup Game** at least once (see `README.md`).

A. Install Windows Build Support (one time)

If you installed Unity through **Unity Hub**:

1. Open **Unity Hub ▶ Installs**.
2. Click the ⚙️ (**gear**) ▶ **Add modules** on your editor version.
3. Tick **Windows Build Support (Mono)** — and optionally (**IL2CPP**) for a faster, harder-to-decompile build.
4. Click **Install** and wait for it to finish.

If you are already on Windows with the standard editor install, this module is usually present.

B. Generate scenes & prefabs (if not done yet)

1. Open the project in Unity.
2. Menu bar ▶ **Space Shooter ▶ Setup Game**.
3. Wait for the “**Setup complete!**” dialog.

This guarantees `Assets/Scenes/MainMenu.unity` and `Assets/Scenes/GameScene.unity` exist and are already added to Build Settings.

C. Configure Build Settings

1. Menu bar ▶ **File ▶ Build Settings...**
2. Under **Scenes In Build**, confirm the list is exactly:
 - `Scenes/MainMenu` → index **0**
 - `Scenes/GameScene` → index **1**

The **MainMenu** must be index **0** so the game boots into the menu.

- Missing scenes? Open each scene, then click **Add Open Scenes**, or re-run **Space Shooter ▶ Setup Game**.

- Wrong order? Drag `MainMenu` to the top of the list.

3. In the **Platform** list on the left, select **Windows, Mac, Linux**.

4. On the right set:

- **Target Platform:** `Windows`

- **Architecture:** `Intel 64-bit (x86_64)`

5. If the platform is not already active, click **Switch Platform** (this may take a minute).

D. (Optional) Player Settings

1. In **Build Settings**, click **Player Settings...** (bottom-left).
 2. Recommended values (the setup script already sets several of these):
 - **Company Name:** IndieDev
 - **Product Name:** Space Shooter
 - **Resolution and Presentation ▶ Fullscreen Mode:** Windowed or Fullscreen Window
 - **Default Screen Width / Height:** 1920 × 1080
 3. (Optional) **Other Settings ▶ Configuration ▶ Scripting Backend:** choose IL2CPP for a more optimized build, or leave Mono for faster build times.
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E. Build

1. Back in **File ▶ Build Settings...**, click **Build** (not Build And Run the first time, so you can inspect the output).
2. Choose/create an output folder, e.g. Builds/Windows .
3. Wait for the build to complete. Unity opens the folder when done.

You will get:

```
Builds/Windows/  
├── Space Shooter.exe           ← run this  
├── UnityPlayer.dll  
├── Space Shooter_Data/        ← game data (keep next to the .exe)  
└── MonoBleedingEdge/          (Mono backend only)
```

F. Run & distribute

1. Double-click **Space Shooter.exe** .
 2. To share the game, zip the **entire** Builds/Windows folder. The .exe will not run without its accompanying _Data folder and UnityPlayer.dll .
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Troubleshooting

Symptom	Fix
“Scene couldn’t be loaded ... not been added to the build settings”	Add both scenes in File ▶ Build Settings (section C).
Menu shows but Play does nothing	Ensure <code>GameScene</code> is in Build Settings at index 1.
Everything is black / no sprites	Run Space Shooter ▶ Setup Game ; sprites are generated at runtime, so make sure scripts compiled with no errors.
No sound	Sound is procedural; check the OS volume and the in-game Pause menu volume sliders.
Buttons not clickable	Confirm an EventSystem exists in the scene (created automatically by the setup script).
Compile errors after import	Use Unity 2021.3 LTS or newer ; older versions lack some C# 8/9 syntax used here.

Enjoy building and shipping **Space Shooter**! 🚀